

# DIPESH SHRESTHA

## Education

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**MASTERS IN DATA SCIENCE** – University of Greenwich – London, UK 2023 – 2024

Majors: Statistical Methods for Time Series Analysis, Applied Machine Learning, Data Visualization, Big Data

Thesis: *Unsupervised Approach to Identify Positive Play in Gambling*

**BACHELORS IN COMPUTER ENGINEERING** – Tribhuvan University – Kathmandu, Nepal 2015 – 2019

Majors: Artificial Intelligence, Simulation & Modeling, Big Data

Thesis: *Sentiment Classification using LSTM for Devnagari Nepali Text*

## Research Experience

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### NEUROPHYSIOLOGICAL MECHANISM OF MENTAL SIMULATION

- Designed and implemented MEG-optimized cognitive tasks in MATLAB to probe mental simulation.
- Developed and tested an analysis pipeline for MEG frequency analysis using Fourier transforms.

### TWO DISTINCT MECHANISMS OF TEMPORAL PRECISION

- Designed two behavioral experiments using Psychopy and Ouvia.
- Collected behavioral data in two phases: online via Prolific, followed by offline experiments including eye tracking.
- Conducted computational modeling and neural data analysis to dissociate sources of variability.
- Created a computational pipeline to quantify the relationship between neural population dynamics and temporal precision.

### EYE MOVEMENTS PREDICT CATEGORY OF ABSTRACT CONCEPTS

- Designed cognitive tasks in MATLAB to decode abstract concepts using eye movements.
- Collected behavioral and eye tracking data in human participants.
- Performed analyses including LDA, time-resolved analysis, and statistical testing.

## Publications

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Neupane, S., Pandey, A., Prasai, S., **Shrestha, D.**, & Neupane, S. (2023). *Developing Science Research by Recognizing the Human Capacity for Inquiry*. Journal of Neuroscience, 43(44), 7243-7246. <https://doi.org/10.1523/JNEUROSCI.1790-23.2023>

Pandey, A., **Shrestha, D.**, & Neupane, S. (2025). *Perceptual-Cognitive Training Paradigms for Understanding Cognition and Cortical Visual Impairments*. In The Hippocampus - Cognitive Functions and Disorders. IntechOpen. <https://doi.org/10.5772/intechopen.1009575>

## Conference Presentation

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**Shrestha, D.**, Eperon, A., Bottini, R., & Neupane, S. (2026). *Two distinct mechanisms of temporal precision*. Poster presentation at the 35th Annual Meeting of Neural Control of Movement, Kobe, Japan.

**Shrestha, D.**, Eperon, A., Bottini, R., & Neupane, S. (2026). *Two distinct mechanisms of temporal precision*. Poster presentation at the Workshop on CONCEPTS, ACTIONS, and OBJECTS, Rovereto (TN), Italy.

## Work Experience

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**RESEARCH ASSISTANT** – Bottini Lab, CIMeC

May 2025 - Present

- Assisted in experimental planning, subject recruitment, and data collection.
- Presented ongoing research and results at lab meetings.

- Actively participated in and contributed to scientific brainstorming and experimental design discussions.

## DATA SCIENCE INTERN – Future Anthem

Jul 2023 - Dec 2023

- Developed a Gaussian Mixture Model in *PySpark*, increasing precision for identifying positive play behavior by **20%**.
- Updated and optimized *Power BI* dashboards using *DAX* enabling real-time monitoring.
- Assisted data scientists with **QA** and model testing ensuring thorough validation and accuracy of models.

## SOFTWARE DEVELOPER – Nimble Clinical Research

Dec 2020 - Dec 2022

- Led the development of a clinical data analysis platform, processing over 50 million clinical records.
- Developed a comprehensive data visualization component, utilizing R programming language and Plotly library.
- Implemented a reporting system using *Jinja* templating engine to generate customizable reports in PDF.

## Skills

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|----------------------------------------------------------|----------------------------------------|
| • SQL (SQL Server, MySQL, PostgreSQL)                    | • Microsoft Azure                      |
| • Python (MNE, Pandas, NumPy, SciPy, Matplotlib, Plotly) | • Web Programming (HTML, CSS, Next JS) |
| • Data Analysis and Modeling                             | • MATLAB ( Psychtoolbox, Brainstorm)   |
| • Microsoft Office Suite                                 | • Microsoft Power BI                   |

## Honours & Awards

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- Awarded **1st** Position in Tech Bihani 3.0 organized by PACE(Project Association for Computer and Electronics) of Advanced College of Engineering and Management.
- Awarded **2nd** Position in KU IT Meet organized by Kathmandu University